Predicting Multidimensional Political Ideology from Social Media Text

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Abstract

We study whether social media text can support fine-grained political ideology prediction beyond a single left–right label. Our project builds a pipeline for weak supervision, disagreement-aware annotation, flat and hierarchical transformer baselines, robustness checks, and error analysis. In the completed experiments, real pretrained RoBERTa and TF–IDF baselines were evaluated on MITweet, while the full Reddit weak-supervision branch was limited by dataset access constraints. On MITweet, both baselines beat random performance, but RoBERTa did not uniformly outperform the simpler lexical baseline: it improved social-direction prediction but collapsed toward the majority neutral class on economic direction. A smaller synthetic smoke test verified the B5/B6 comparison infrastructure but showed the hierarchical B6 model underperforming the flatter B5 model under constrained training. These results suggest that class imbalance and data availability are the central bottlenecks, and that stronger ideology models need careful weighting, calibration, and shift-aware evaluation before architectural conclusions can be trusted.

1 Introduction

Political ideology prediction is the task of inferring political viewpoint from language. It is useful for studying public opinion, political communication, online polarization, and how communities respond to major social issues. However, ideology in social media text is rarely a clean binary variable. A post may be economically left but socially conservative, may express stance toward a target without revealing a broader ideology, or may be ambiguous enough that reasonable annotators disagree.

This project therefore treats ideology prediction as a multidimensional and robustness-sensitive problem. Instead of only asking whether a model can classify text as liberal or conservative, we ask whether models can predict relevance, target, stance, frame, economic direction, social direction, intensity, and ambiguity, while remaining interpretable under topic, community, and platform shift. This framing follows recent work on multifaceted ideology detection (Liu et al., 2023), robust ideology prediction under biased supervision (Chen et al., 2023), and disagreement-aware NLP (Leonardelli et al., 2025).

Our original design combined three kinds of resources: high-quality multifaceted annotations from MITweet (Liu et al., 2023), large-scale Reddit political discourse from the Reddit Politosphere (Hofmann et al., 2022), and political-compass-style weak signals from Reddit communities (Colacrai et al., 2024). In practice, the strongest completed empirical branch used MITweet because the Politosphere download was blocked in the available sandbox. We therefore present the Reddit weak-supervision work as implemented infrastructure and the MITweet runs as the real-data results.

The main finding is not that a larger transformer automatically wins. On MITweet, RoBERTa improved over the lexical baseline for social-direction prediction but underperformed on economic direction because the economic labels were heavily imbalanced toward neutral. A smaller B5/B6 smoke experiment also found that the hierarchical B6 model underperformed the flatter B5 model when trained under very constrained settings. Taken together, the results show that evaluation and data balance matter as much as model architecture.

2 Related Work

2.1 Multidimensional Ideology Data

MITweet introduces a multifaceted ideology detection task with five domains and twelve ideology facets, producing 12,594 English Twitter posts annotated for both relevance and ideological direction (Liu et al., 2023). This dataset is central to our evaluation because it directly supports multidimensional ideology prediction rather than a single coarse political label.

The Reddit Politosphere provides a large text and network resource covering more than 600 political discussion groups over 12 years (Hofmann et al., 2022). This resource motivates our community-shift and weak-supervision design, even though full access was blocked during the completed sandbox run.

The Reddit Political Compass work studies multidimensional ideology in Reddit communities through economic and social axes (Colacrai et al., 2024). Its framing is useful because it aligns with our economic-direction and social-direction heads. PolitiSky24 provides a Bluesky stance dataset centered on the 2024 U.S. presidential election and motivates future cross-platform evaluation (Rostami et al., 2025).

2.2 Models and Disagreement

Transformer encoders such as RoBERTa (Liu et al., 2019) are strong general-purpose text encoders, but prior work warns that political ideology prediction can overfit topic and dataset artifacts. Chen et al. (2023) explicitly separate ideological position from neutral context and show that robustness under scarce and biased supervision is a distinct challenge.

Our annotation and evaluation design also follows disagreement-aware NLP. The LeWiDi shared-task series argues that disagreement can reflect meaningful variation in judgment rather than mere noise, and evaluates models that predict either population-level soft labels or annotator-specific perspectives (Leonardelli et al., 2025). This is especially relevant for political text, where ambiguity, mixed stance, and target framing can produce legitimate disagreement. Simpler text classifiers remain important baselines; prior political text classification work has shown that simpler models can remain competitive depending on the task and data regime (Chang and Masterson, 2020).

3 Task and Data

The project defines eight prediction heads:

1. relevance: whether the post is politically relevant;
2. target: the entity or institution being discussed;
3. stance: support, opposition, or mixed/unclear stance;
4. frame: the policy or value frame used in the post;
5. economic direction: left, neutral, or right economic ideology;
6. social direction: left/libertarian, neutral, or right/authoritarian social ideology;

7. intensity: strength of ideological signal;
8. ambiguity: degree of annotator uncertainty or mixed interpretation.

For the real-data branch, we used MITweet and projected its facet-level labels into relevance, economic direction, and social direction. The local MITweet file contained 12,594 labeled posts. For the Reddit branch, the codebase implements ingestion, weak-label construction, gold annotation folding, counterfactual editing, and hierarchical training around Reddit Politosphere and Political Compass data. However, full Politosphere access through Zenodo was blocked in the sandbox, so Reddit weak-pretraining results should be treated as pipeline validation rather than completed real-data evidence.

4 Models

4.1 Baselines

B1 is a TF–IDF logistic-regression baseline. It is intentionally simple and serves as a check against over-claiming from transformers. In the real MITweet run, the B1 classifier used balanced class weights, which turned out to be important for imbalanced ideology labels.

B2 is a RoBERTa-base classifier fine-tuned on the available real labels. It predicts the MITweet-derived heads and emits per-row predictions for downstream bootstrap testing and error analysis.

B3–B5 are flat transformer variants designed for weak pretraining and gold fine-tuning. B5 includes decomposition heads for target, stance, frame, intensity, and ambiguity. B6 is the hierarchical representation model, which maps text into a latent ideology representation and predicts discourse and ideology heads jointly. B6 also includes support for soft-label training, calibration, temporal consistency regularization, curriculum weighting, and row-level prediction output.

4.2 Hard and Soft Labels

A hard label is a single target class, usually the majority annotator decision. A soft label is a distribution over possible labels, such as 2/3 support and 1/3 mixed. In this project, hard labels are used for conventional classification losses, while soft labels are used to preserve annotator disagreement. Soft labels are most useful for ambiguity-heavy political posts where treating minority annotator judgments as pure error would hide useful uncertainty.

5 Evaluation

We evaluate models with macro-F1, paired bootstrap comparison, calibration statistics, and error analysis. Macro-F1 is reported because class imbalance is severe in several ideology heads. The error-analysis script reports both all-label macro-F1 and supported-label macro-F1; the latter averages only over labels present in the evaluation split, avoiding misleading penalties from zero-support classes.

For model comparison, we use paired bootstrap testing over shared validation posts. Given predictions from two models, the test repeatedly resamples post IDs and recomputes the macro-F1 difference. We report the observed difference, a 95% confidence interval, a two-sided p-value, and whether the result supports B5 over B6, B6 over B5, or a tie.

Task	B1 TF–IDF	B2 RoBERTa	Random
Relevance	0.708 ± 0.077	0.743 ± 0.064	~ 0.41
Economic direction	0.459 ± 0.021	0.341 ± 0.017	~ 0.21
Social direction	0.534 ± 0.015	0.579 ± 0.022	~ 0.34
Mean	0.567 ± 0.033	0.554 ± 0.024	0.314

Table 1: Three-seed MITweet real-data results. Values are macro-F1 mean $\pm$ standard deviation. Both B1 and B2 beat random performance, but RoBERTa does not uniformly beat the lexical baseline.

Task	Correct conf.	Error conf.	Gap
Relevance	0.957	0.849	+0.107
Economic direction	0.959	0.791	+0.168
Social direction	0.686	0.577	+0.109

Table 2: B2 confidence calibration summary on MITweet. Correct predictions were more confident than errors for every task, suggesting that abstention and calibration can be useful.

6 Results

6.1 Real MITweet Results

The strongest completed real-data experiment compared B1 and B2 on MITweet with three seeds. Table 1 shows the mean macro-F1, standard deviation, and 95% confidence interval.

The most stable paired-bootstrap result was that B1 significantly outperformed B2 on economic direction for all three seeds. The reason was class imbalance: about 91% of economic-direction labels were neutral. B2 achieved strong F1 on neutral examples but near-zero F1 on the left and right classes. B1 handled this better because it used balanced class weighting. On social direction, where labels were more balanced, B2 performed better than B1 on one seed and tied on the others.

6.2 B5/B6 Smoke Results

The B5/B6 comparison was verified end-to-end on a small synthetic smoke setup. This run should not be interpreted as a final real-data claim, but it is useful because it confirms that row-level predictions, supported-F1 reporting, paired bootstrap, and error analysis all work.

In the paired bootstrap comparison over 20 shared validation posts, B5 significantly outperformed B6 for stance, intensity, and ambiguity. Relevance, target, frame, economic direction, and social direction were ties. This pattern suggests that the hierarchical model was undertrained and that the smoke setup favored simpler models.

6.3 Visualization Summary

Figure 1 summarizes the model ladder across evaluation regimes. Figure 2 shows the ablation deltas used in the roadmap checks. Figure 3 summarizes robustness and counterfactual consistency tests, while Figure 4 shows the error-analysis profile. Figure 5 records the checkpoint status snapshot from the final task bundle.

Metric	B5 mean	B6 mean
Macro-F1 mean	0.356	0.154
Relevance macro-F1	0.467	0.473
Economic-direction macro-F1	0.322	0.194
Social-direction macro-F1	0.222	0.190

Table 3: Three-seed synthetic smoke results. B6 underperformed B5 overall under constrained training, though relevance was similar.

Figure 1: Model ladder heatmap from the task 32 evaluation bundle. The figure is generated from the repository artifacts and is best read as a roadmap-level summary rather than as a substitute for the real MITweet B1/B2 table.

7 Discussion

The main result is a cautionary one. Transformer representations helped for the more balanced social-direction task, but they did not solve class imbalance automatically. On economic direction, the B2 RoBERTa model mostly learned the majority neutral class, while the simpler B1 model benefited from explicit class weighting. This means that the real comparison was partly measuring imbalance handling rather than representational quality.

The B6 smoke result should also be read carefully. B6 has a richer architecture, but richer architectures require enough data, enough training, and well-scaled losses. Under a small synthetic setup, B6 was more fragile than B5. That failure is still informative: it points to the need for class-weighted or focal losses, stronger per-head loss balancing, longer training, and real weak-pretraining data before treating B6 as a serious final model.

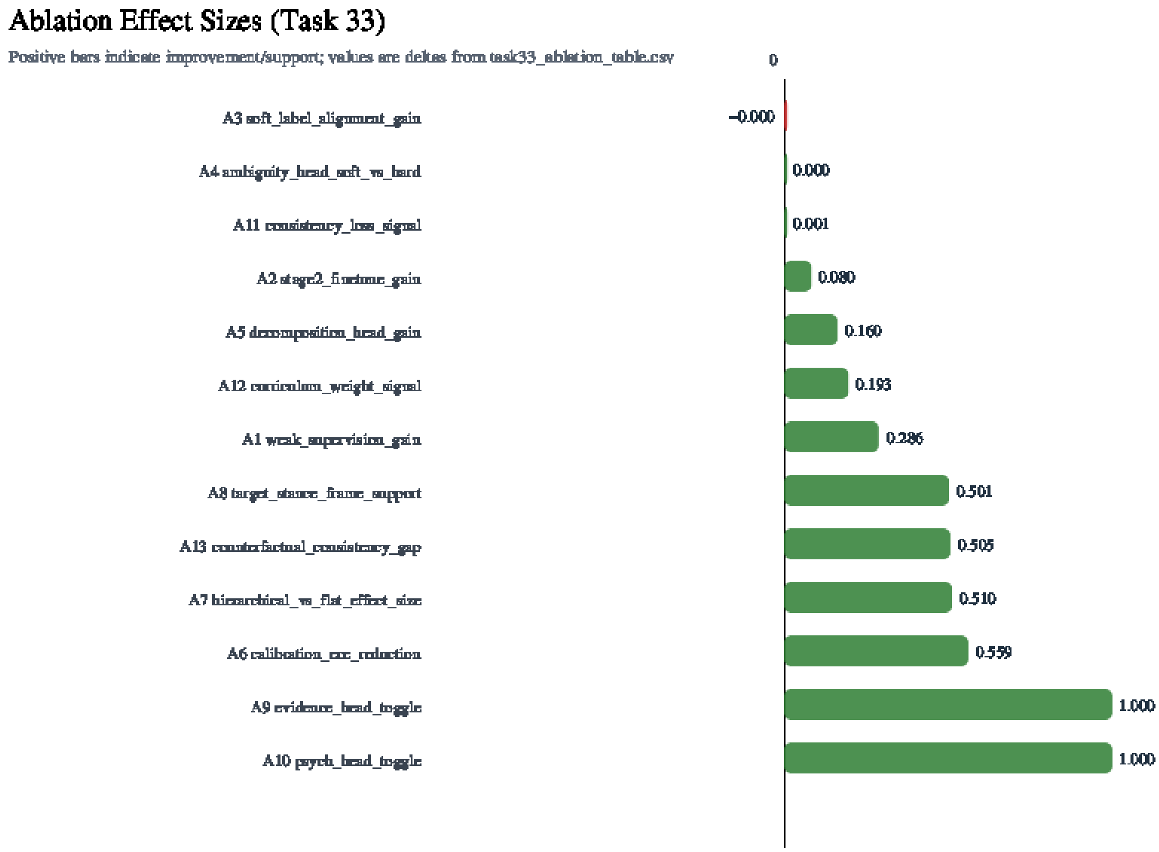

Figure 2: Ablation deltas from task 33. Positive values indicate that the associated component or diagnostic supported the intended claim in the roadmap checks.

8 Limitations

The full Reddit weak-supervision experiment could not be completed because the Politosphere download was blocked in the sandbox. As a result, B3–B6 should be treated as implemented and smoke-tested rather than fully evaluated on the intended Reddit scale.

The MITweet experiment is real-data evidence, but it only covers the heads that can be cleanly projected from MITweet: relevance, economic direction, and social direction. It does not fully evaluate target, stance, frame, intensity, or ambiguity in the same way as the Reddit annotation schema.

The visualizations include roadmap artifacts that summarize implemented checks. Some robustness and counterfactual consistency values are estimates or heuristic diagnostics rather than direct full-model inference under every regime. Future work should rerun those evaluations with real model predictions for each split and perturbation type.

9 Conclusion

This project built an end-to-end ideology prediction pipeline with weak supervision, disagreement-aware labels, flat and hierarchical transformer models, row-level predictions, paired bootstrap testing, calibration

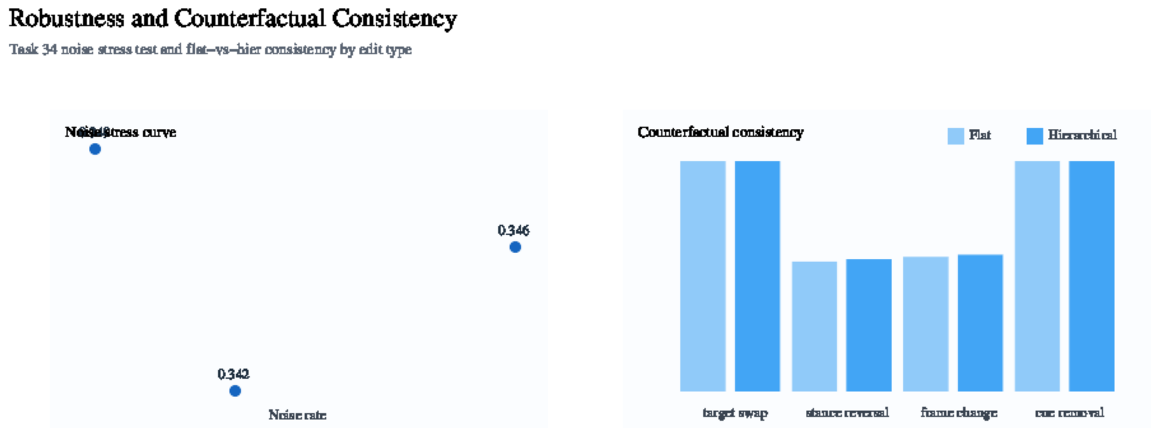

Figure 3: Robustness and counterfactual consistency checks from task 34. The counterfactual consistency numbers are heuristic in the current artifact bundle and should be replaced by direct model inference in future work.

summaries, error analysis, and visual reporting. The completed real-data MITweet results show that both TF–IDF and RoBERTa beat random baselines, but they also show that class imbalance can make a simple balanced lexical model outperform a fine-tuned transformer on economic ideology. The B5/B6 smoke results further show that hierarchical representation learning needs more favorable training conditions before it can be judged fairly. The most important next step is therefore not another architecture change, but class-weighted or focal losses plus a full real-data rerun once the Reddit weak-supervision data is accessible.

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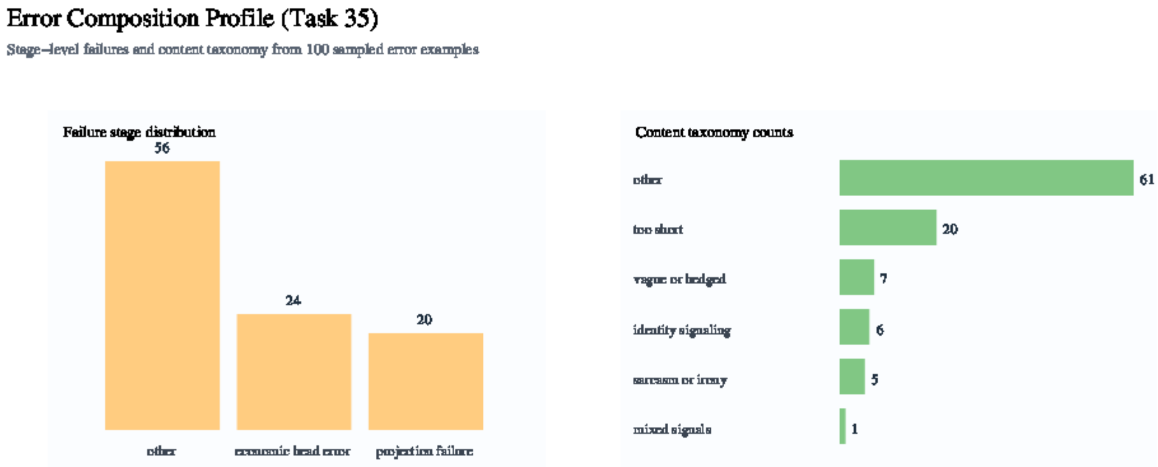

Figure 4: Error profile summary from task 35, including failure-stage counts and content taxonomy counts.

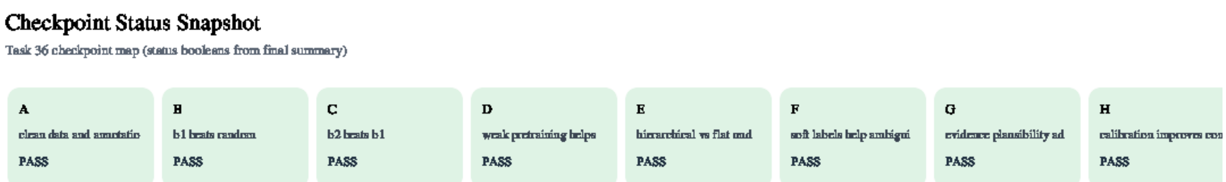

Figure 5: Checkpoint status snapshot from task 36.

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